Information and decoherence in a muon-nuclei coupled system
Supplemental information

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A. μSR experiments

In the muon experiment, a beam of spin-polarized muons were incident on a sample, and the number of positrons detected in both the forwards and backwards detectors, $N_F(t)$ and $N_B(t)$ respectively, was measured [1]. The muon asymmetry was calculated as

$$A(t) = \frac{N_B(t) - \alpha N_F(t)}{N_B(t) + \alpha N_F(t)}, \tag{1}$$

where the parameter α takes into account systematic differences between the readings of both sets of detectors. Our experiments were performed using the MuSR spectrometer at the ISIS Facility, Rutherford Appleton Laboratory, UK. Polycrystalline samples of CaF_2 and NaF were wrapped in sheet of 25 μm silver foil, and were placed in Variox cryostats, kept at temperatures of 50 K and 100 K respectively in zero applied magnetic field. The Earth's magnetic field was compensated to better than 50 μT using active field compensation. The mean muon lifetime is 2.2 µs, but data can be obtained out to at least ten times this value at ISIS if collected for several hours. Around 181 million and 720 million muon decay events were detected for CaF_2 and NaF respectively. These statistics are much higher than are normally taken for muon data, highlighting the importance of high data rate acquisition of time-differential data at pulsed sources. High statistics are needed to resolve long-time features which are important in revealing the details of the more distant couplings.

B. DFT+μ calculations

The ab initio calculations were performed with the QUANTUM ESPRESSO package [2]. The calculations were performed in a supercell containing $2\times2\times2$ conventional unit cells. For the diamagnetic states considered here, the +1 charge state of the muon was determined by the charge of the supercell. A muon was placed in several randomly chosen low-symmetry sites and all ions were allowed to relax until the forces on all ions and the energy change had fallen below a convergence threshold.

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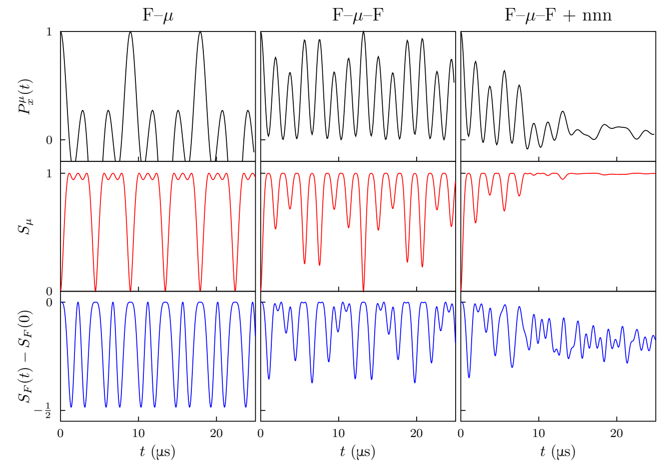
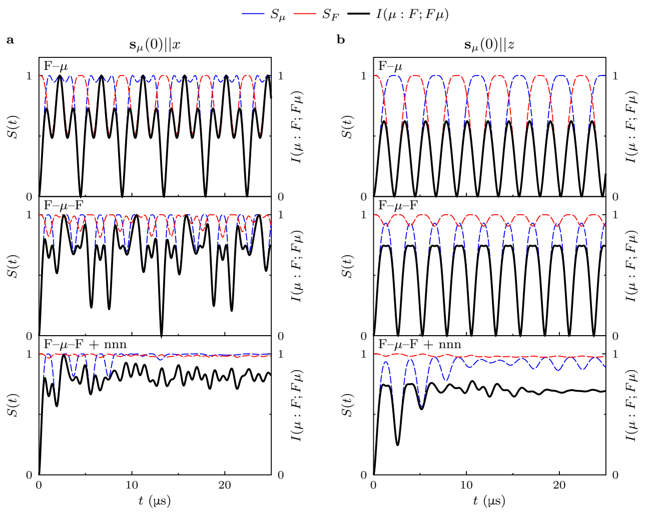


Supplementary Fig. 1. **von Neumann entropy for muon-fluorine states.** The F–μ–F (or F–μ) axis is aligned along z and the implanted muon spin $\boldsymbol{s}_\mu$ is initially aligned along x. For the three cases of isolated F–μ, isolated F–μ–F and environmentally decohered F–μ–F, the plots show the time-dependence of the muon polarization $P_x^\mu(t)$, the muon entropy S_μ (obtained by tracing out all the other spins), and the entropy of the entire fluorine system, S_F.

C. The entropy in F–μ–F state for different orientations

To gain a proper understanding of the dissipation of information within the CaF_2 crystals, the time evolution of the entropy with the muon polarisation initially perpendicular to the F–μ–F bond, that is $\boldsymbol{s}_\mu(0)||x$, was calculated. This is shown in Supplementary Fig. 1. This figure, like the case for the muon initially polarized along z (described in the main paper), shows that the introduction of next-nearest neighbour (nnn) fluorine nuclei causes the muon to lose its polarization, and hence information as it evolves into a mixed state.

Further insight can be obtained by investigating the mutual information between the fluorines and the muon. This is plotted in Supplementary Fig. 2. The mutual information is used here as a measure of the extent of the correlations between the muon and the fluorines. One can see that the muon becomes correlated with environment (due to being entangled with the fluorine nuclei), which causes the quantum information to be decohered into the environment, and hence the expectation value of the muon's polarisation tends to decrease as measured.



Supplementary Fig. 2. **Mutual information in the F–μ system. a**, The mutual information in the F–μ, F–μ–F, and F–μ–F state with the next-nearest neighbour (nnn) couplings included, where the initial muon polarization is perpendicular to the F–μ bond. Here, S_μ is the entropy of the muon, and S_F is the entropy of all the N fluorine nuclei included in the calculation. **b**, the same as **a** but with the initial muon polarization parallel to the F–μ bond.

One can gain some insight into the timescale over which the decoheres via a very rough estimate. This timescale is given *very roughly* by the spacing between the original F–μ–F energy levels ($\sim \hbar\omega_{\rm D}$) divided by the broadening of these levels due to the couplings to the next-nearest neighbours ($\sim \hbar\omega_{\rm D}/4$) in units of the period of the oscillations. Thus the oscillations die away in approximately four periods $\sim 10^{-5}$ s.

D. Calculation of the muon polarization

As stated in the main paper, the the time evolution of the muon's spin (labelled here as spin $i=0$) is given by

$$P^\mu(t) = \frac{1}{2}\left\langle \mathrm{Tr}\left[\boldsymbol{\sigma}^\mu_{\hat{n}} \exp\left(\frac{-\mathrm{i}\hat{\mathcal{H}}t}{\hbar}\right)\boldsymbol{\sigma}^\mu_{\hat{n}} \exp\left(\frac{\mathrm{i}\hat{\mathcal{H}}t}{\hbar}\right)\right]\right\rangle_{\hat{n}}, \quad (2)$$

where $\langle\ldots\rangle_{\hat{n}}$ represents the angular average over $\hat{n}$ (appropriate for an experiment on a polycrystalline sample), and $\boldsymbol{\sigma}^\mu_{\hat{n}}$ is the Pauli spin operator for the muon in the direction of $\hat{n}$. This results in an oscillatory time-dependent polarization of the general form

$$P^\mu(t) = \sum_{mn} a_{mn}\cos(\omega_m - \omega_n)t, \quad (3)$$

where the sum is taken over all pairs of energy levels (see for example [3]). The amplitude a_{mn} of each term in this sum is proportional to $|\langle m|\,\sigma^\mu_x\,|n\rangle|^2 + |\langle m|\,\sigma^\mu_y\,|n\rangle|^2 + |\langle m|\,\sigma^\mu_z\,|n\rangle|^2$ and as a result most possible oscillations are present in the spectrum.

Because the time-dependent muon polarization and reduced entropy of the muon involve quantum interference terms between every pair of energy eigenstates, and since the differences in energy between different pairs of levels appear in complicated ratios with each other as the number of spins included increases, the muon will never return to a pure state even if one could measure well beyond the current limit of ten to fifteen muon lifetimes. The precise value of the couplings also depend on the distortions introduced by the muon, pulling in the nearest-neighbour fluorines, repelling the nearest cations etc, and the size of these distortions depends on the energetics of the system in a complicated way. (In fact, even in the absence of these distortions, and ignoring more distant dipolar couplings and even the coupling between the two fluorines in the F–μ–F system, the simple model of F–μ–F in Ref. 4 contains a mixture of frequencies in the ratio $\sqrt{3}$:$\frac{3-\sqrt{3}}{2}$:$\frac{3+\sqrt{3}}{2}$ and these are in an irrational ratio and so the oscillatory behaviour never repeats perfectly.)

E. Contributions to the second moment of the dipolar field

The second moment of the dipolar field contains a term proportional to $\sum_{j=1}^{\infty} 1/r_j^6$. For the CaF_2 system, the fluorine ions are arranged on a simple cubic lattice with lattice constant $a/2$. The muon site is at $0\,\frac{1}{4}\,0$ in fractional coordinates, and so has 2 nearest-neighbour fluorines, 8 next-nearest-neighbour fluorines, 10 next-next-nearest neighbour fluorines and so on. Thus

$$\sum_{j=1}^{\infty}\frac{1}{r_j^6} = \left(\frac{4}{a}\right)^6 \Sigma_\infty, \quad (4)$$

where

$$\Sigma_\infty = \frac{2}{1^6} + \frac{8}{(\sqrt{5})^6} + \frac{10}{3^6} + \ldots = 2.0908748, \quad (5)$$

where the final result is obtained by summing the terms out to some distance R (in units of $a/4$) and then evaluating the remaining sum from R to ∞ by expressing the sum as an integral which gives a term $\pi/[6R^3]$. Note that the first term in this series is 2, while the second term is $\frac{8}{125} = 0.0185$, demonstrating the rapid convergence of this series. The sum to infinity of all the terms after the first are only a factor $\zeta^{-6} \equiv (\Sigma_\infty - 2)/\frac{8}{125} = 1.41992$ bigger than the second term. This gives $\zeta = 0.943$ and hence scaling the nnn interaction with ζ using

$$\sigma^2_\infty = \sigma^2_{\rm nn} + \frac{2}{3}\left(\frac{\mu_0}{4\pi}\right)^2\hbar^2\gamma_\mu^2 \sum_{j\in{\rm nnn}} \frac{\gamma_j^2 I_j(I_j+1)}{(\zeta r_j)^6}, \quad (6)$$

which is equation (2) of the main paper, allows the nnn coupling to act as a proxy for all more distant couplings in the lattice. These expressions do not include the distortions in the nearby fluorine ions due to the muon.

Including these, obtained from our DFT+μ calculations, decreases ζ to the value 0.937.

A similar analysis can be performed for NaF which adopts the rocksalt structure. We now use eqn 4 with

$$\Sigma_\infty = \frac{2}{(\sqrt{2})^6} + \frac{4}{(\sqrt{6})^6} + \frac{4}{(\sqrt{10})^6} + \ldots = 0.2792433, \quad (7)$$

where the sum has been evaluated using the same method as for CaF_2. This case is more complicated because there are two contributions to the second moment, one from the fluorine nuclei, ($I = \frac{1}{2}$), and one from the sodium nuclei ($I = \frac{3}{2}$).

F. Quadrupolar interactions in NaF

Sodium nuclei have a spin of $I = \frac{3}{2}$, and as such they are susceptible to the quadrupolar as well as the dipolar interaction. This was taken into account by adding the additional term $\hat{\mathcal{H}}_Q$ to the Hamiltonian

$$\hat{\mathcal{H}}_Q = \sum_{i \in \mathrm{Na}} \frac{eQ(1+\gamma)}{\hbar 2I(2I-1)} \sum_{\alpha\beta} V^i_{\alpha\beta} \hat{I}^i_\alpha \hat{I}^i_\beta, \quad (8)$$

where

$$\begin{aligned} V^i_{\alpha\beta} &= \frac{\partial^2 V(\boldsymbol{x}^i)}{\partial x_\alpha \partial x_\beta} \quad (9) \\ &= \sum_{j \neq i} \frac{q_j}{4\pi\epsilon_0} \left(\frac{3(x^i_\alpha - x^j_\alpha)(x^i_\beta - x^j_\beta) - \delta_{\alpha\beta}|\boldsymbol{x}^i - \boldsymbol{x}^j|^2}{|\boldsymbol{x}^i - \boldsymbol{x}^j|^5} \right), \end{aligned}$$

is the external electric field gradient (EFG) at the site of the ith quadrupolar (Na) nucleus (with spin operator $\boldsymbol{I}^i$), due to charges q_j, and the sum is performed over all atoms in the lattice. The tensor is traceless and so $V^i_{xx} + V^i_{yy} + V^i_{zz} = 0$. For Na, the quadrupole coupling factor $Q = 0.10 \times 10^{-28}$ m^2, and the anti-shielding factor γ is 4 [5].

In pure NaF there is no EFG at the Na site since all contributions from other ions cancel by symmetry, but the presence of the muon leads to three effects which result in a non-zero EFG at the two Na sites closest to the muon.

1. The presence of the muon itself leads to an EFG at each Na site resulting from the muon's charge of $+e$.

2. The displacement of each Na from its equilibrium position due to the implanted muon means that the EFG contributions from the other ions in the crystal no longer cancel.

3. A third, smaller effect, is due to the other muon-induced local distortion of ions (principally the two fluorine ions adjacent to the muon and the other sodium cation).

We included all these contributions in our calculation of the EFG tensor $V^i_{\alpha\beta}$ and added $\hat{\mathcal{H}}_Q$ (eqn 8) into our Hamiltonian which was then used for evaluating the time dependence of the density matrix and hence the muon polarization. Including the quadrupole interaction is important in obtaining a good fit at long times and is therefore particularly needed for our very high statistics data.

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